

Program : Diploma in Printing Technology	
Course Code : 6102D	Course Title: Disaster Management (PT)
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P: 0)	Periods per semester: 60

Course Objectives:

- To learn about various types of natural and man-made disasters.
- To know pre- and post-disaster management for some of the disasters.
- To know about various information and organizations in disaster management in India.
- To get exposed to technological tools and their role in disaster management.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Causes of flood, drought, forest fire, deforestation, global warming & various pollutions.		Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Acquainted with basic information on various types of disasters and to knowing the precautions and awareness regarding various disasters.	15	Understanding
CO2	Decide action to be taken under various disasters.	14	Understanding
CO3	Familiarized with organizations in India which are dealing with disaster management.	15	Understanding
CO4	Able to adopt Scientific and Technological approach for Disaster Management.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						3
CO3	3						3
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Acquainted with basic information on various types of disasters and to knowing the precautions and awareness regarding various disasters.		
M1.01	Understanding the Concepts of Disaster.	2	Understanding
M1.02	Geological Disasters.	2	Understanding
M1.03	Hydro-Meteorological Disasters.	2	Understanding
M1.04	Biological Disasters.	2	Understanding
M1.05	Technological Disasters and Manmade Disasters.	3	Understanding
M1.06	Global Disaster Trends and Risk.	2	Understanding
M1.07	Urban Disasters.	2	Understanding
Content: Understanding the Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, and Capacity. Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro - Meteorological Disasters (floods, cyclones, lightning, thunder - storms, hail storms, avalanches, droughts, cold and heat waves); Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear); Manmade Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters); Global Disaster Trends - Emerging Risks of Disasters - Climate Change and Urban Disasters.			
CO2	Decide first action to be taken under various disasters.		
M2.01	Disaster and Development and disaster management.	3	Understanding
M2.02	Disaster Management Cycle	2	Understanding
M2.03	Pre-Disaster.	2	Understanding
M2.04	Prevention and Mitigation of Disasters.	3	Understanding
M2.05	Awareness during Disaster.	2	Understanding

M2.06	Post-disaster.	2	Understanding
	Series Test I	1	
Contents: Disaster and Development, and disaster management-Introduction. Disaster Management Cycle - Paradigm Shift in Disaster Management. Pre-Disaster - Risk Assessment and Analysis, Risk Mapping, zonation and Micro zonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development. Awareness during Disaster - Evacuation - Disaster Communication - Search and Rescue - Emergency Operation Centre - Incident Command System - Relief and Rehabilitation - Post-disaster - Damage and Needs Assessment, Restoration of Critical Infrastructure - Early Recovery - Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action.			
CO3	Familiarized with organizations in India which are dealing with disaster management.		
M3.01	Disaster Profile of India.	3	Understanding
M3.02	Disaster Management Act 2005.	3	Understanding
M3.03	Role of Government (local, state and national)	5	Understanding
M3.04	Non-Government and Inter Governmental Agencies.	4	Understanding
Contents: Disaster Profile of India - Mega Disasters of India and Lessons Learnt. Disaster Management Act 2005 - Institutional and Financial Mechanism, National Policy on Disaster Management, National Guidelines and Plans on Disaster Management. Role of Government (local, state and national), Non-Government and Inter Governmental Agencies.			
CO4	Able to adopt Scientific and Technological approach for Disaster Management		
M4.01	Geo-informatics in Disaster Management.	3	Understanding
M4.02	Disaster Communication System.	3	Understanding
M4.03	Land Use Planning and Development Regulations.	3	Understanding
M4.04	Structural and Non-Structural Mitigation of Disasters.	3	Understanding
M4.05	S&T Institutions for Disaster Management in India.	2	Understanding
	Series Test II	1	
Contents: Geo-informatics in Disaster Management (RS, GIS, GPS and RS). Disaster Communication System (Early Warning and Its Dissemination). Land Use Planning and Development Regulations, Disaster Safe Designs and Constructions, Structural and Non-Structural Mitigation of Disasters. S&T Institutions for Disaster Management in India.			

Text / Reference:

T/R	Book Title/Author
T1	Publications of National Disaster Management Authority (NDMA) on Various Templates and Guidelines for Disaster Management.
R1	Bhandani, R. K., An overview on natural & man-made disasters and their reduction , CSIR, New Delhi
R2	Srivastava, H. N., and Gupta G. D., Management of Natural Disasters in developing countries. Daya Publishers, Delhi
R3	Alexander, David, Natural Disasters , Kluwer Academic London
R4	Ghosh, G. K., Disaster Management , A P H Publishing Corporation
R5	Murthy, D. B. N., Disaster Management: Text & Case Studies , Deep & Deep Pvt. Ltd.

Online resources:

Sl. No	Website Link
1	https://www.iwapublishing.com/news/disaster-management
2	https://en.wikipedia.org/wiki/Disaster_management_in_India
3	https://ndma.gov.in/
4	https://www.physio-pedia.com/Disaster_Management